

Energy-Saving Fan and Light Automation System

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1 Introduction

Modern smart homes require automated systems that can intelligently control appliances to save energy. Lights and fans are two of the most commonly used electrical devices, and improper manual usage often leads to unnecessary power consumption. With the growing emphasis on energy conservation and sustainable living, there is an urgent need for intelligent automation solutions that can optimize electricity usage while maintaining user comfort.

This project presents an Energy-Saving Fan and Light Automation System using Arduino Uno, where:

- LDR sensor controls bulb brightness based on ambient light
- TMP36 sensor controls fan speed based on temperature

The system employs Pulse Width Modulation (PWM) technique to precisely control the intensity of light and speed of fan, ensuring optimal energy utilization. By continuously monitoring environmental parameters and automatically adjusting appliance operations, the system eliminates human intervention while providing a comfortable living environment. This approach not only reduces electricity bills but also contributes to environmental sustainability by minimizing carbon footprint.

The system works autonomously and provides efficient energy usage without human intervention, making it an ideal solution for residential, commercial, and educational settings where energy conservation is a priority.

2 System Overview

The system continuously monitors environmental conditions:

2.1 Lighting Control

- When the room becomes darker, the LED bulb becomes brighter
- When the room is bright, LED brightness decreases automatically

2.2 Temperature Control

- When temperature rises above 25°C, fan begins to run
- Fan speed increases gradually up to 30°C

2.3 Key Features

- Fully automatic operation
- PWM-based smooth brightness and speed control
- Low power and low-cost components
- Suitable for real-time smart home deployment

2.4 Architecture Diagram

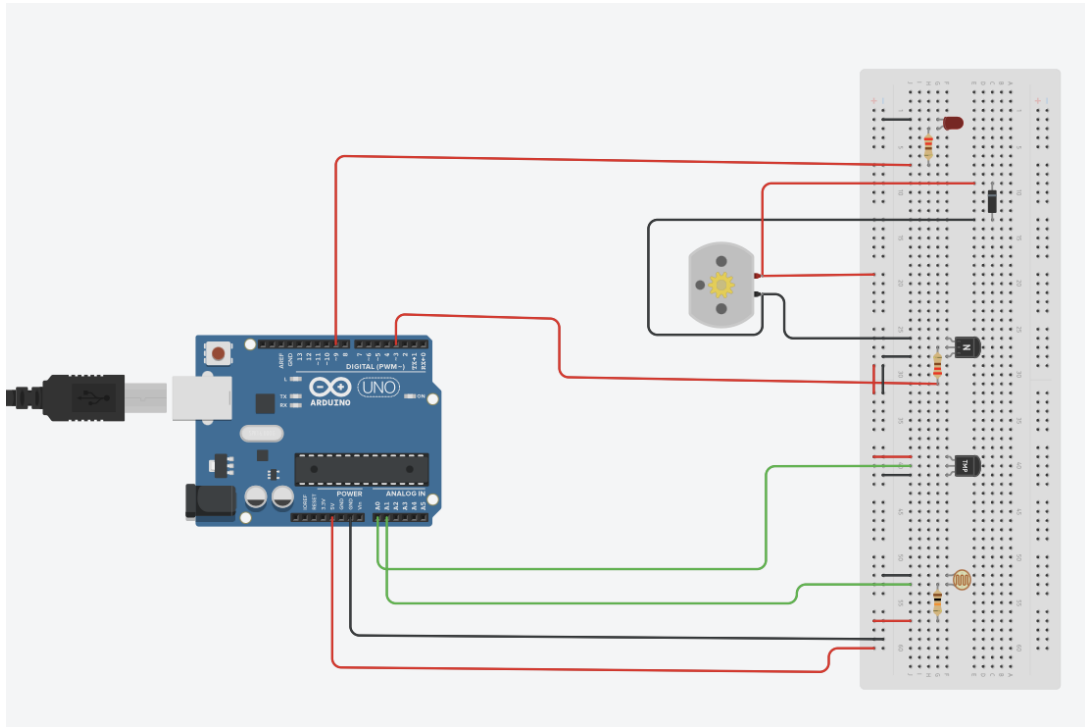


Figure 1: System Architecture Diagram

3 Real Model Image

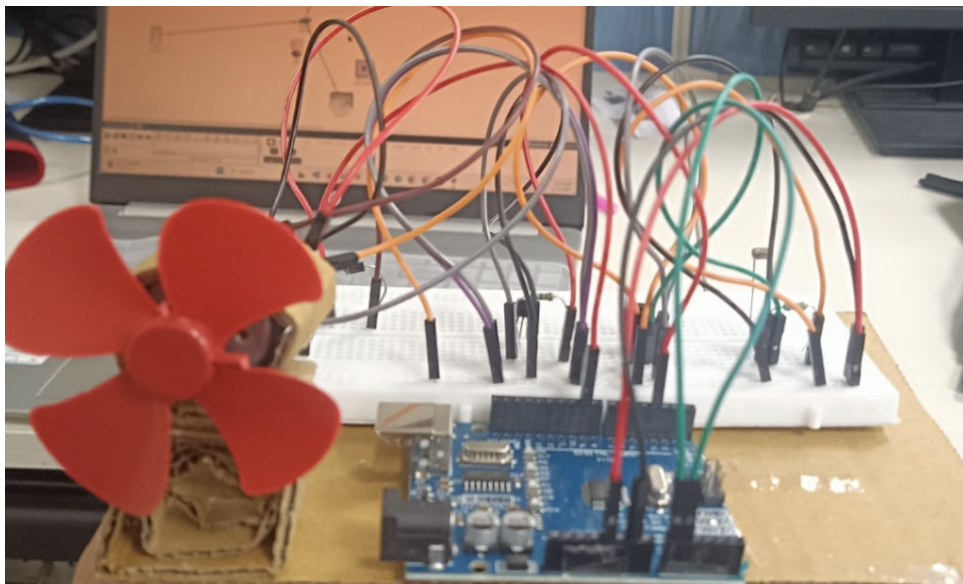


Figure 2: Physical Implementation of the System

4 Hardware Requirements

Component	Specification	Purpose
Microcontroller	Arduino Uno R3	Main controller
Temperature Sensor	TMP36	Temperature monitoring
Light Sensor	LDR with 10k resistor	Ambient light detection
DC Motor	Fan	Air circulation
LED Bulb	with 220 resistor	Lighting
Transistor	NPN (2N2222 / BD139)	Motor control
Diode	1N4007	Motor protection
Breadboard	Solderless	Circuit prototyping

Table 1: Hardware Components Specification

5 Software Requirements

Software	Purpose
Arduino IDE	Code development and uploading
Serial Monitor (9600 baud)	Real-time monitoring and debugging
Embedded C (Arduino)	System programming

Table 2: Software Requirements

6 Deployment

6.1 Step 1: Hardware Setup

- Connect TMP36 → A0
- Connect LDR with voltage divider → A1
- Connect motor via transistor → D3
- Connect LED with resistor → D9
- Connect 5V and GND properly

6.2 Step 2: Upload Software

- Select Board: Arduino Uno
- Select Port: COMx
- Upload the provided code

6.3 Step 3: Run System

- Observe Serial Monitor for temperature and LDR values
- Increase light $\rightarrow$ LED dims
- Decrease light $\rightarrow$ LED brightens
- Increase temperature $\rightarrow$ Fan speeds up

7 Operations

7.1 Lighting Operation

- In bright conditions: Bulb brightness reduces
- In dark conditions: Bulb brightness increases

7.2 Fan Operation

- Below 25°C $\rightarrow$ Fan off
- $25\text{--}30^{\circ}\text{C}$ $\rightarrow$ Speed increases gradually using PWM
- Above 30°C $\rightarrow$ Fan at max speed

7.3 Energy Efficiency

- Components operate only when needed
- Reduces electricity wastage
- Ideal for smart home automation

8 Testing and Results

8.1 Performance Testing

Test Case	Temperature ($^{\circ}\text{C}$)	LDR Value	Fan Speed (PWM)	Bulb Brightness
Normal Day	24	100	0	0
Cloudy Day	25	300	0	85
Warm Evening	27	500	102	170
Hot Night	29	700	204	212
Extreme Condition	31	900	255	255

Table 3: System Response under Different Environmental Conditions

8.2 Energy Savings Analysis

Scenario	Traditional System	Our System	Savings
Daylight Hours	100% brightness	0-30% brightness	70-100%
Cool Weather	100% fan speed	0% fan speed	100%
Night Time	100% brightness	70-100% brightness	0-30%

Table 4: Energy Consumption Comparison

9 Code Implementation

```
// -----  
// PROJECT: ENERGY-SAVING FAN + LIGHT SYSTEM  
// AUTHOR: SATYAM BHARDWAJ (NIT Patna)  
// COMPONENTS: TMP36 (A0), LDR (A1), DC Motor (D3), LED (D9)  
// -----  
  
const int TMP36_PIN = A0; // Temperature sensor (TMP36)  
const int LDR_PIN   = A1; // Light sensor (LDR)  
const int FAN_PIN   = 3;  // DC Motor (Fan) - PWM control  
const int BULB_PIN  = 9;  // LED (Light) - PWM control  
  
// Temperature range for fan speed control  
const float MIN_TEMP = 25.0; // Minimum temperature (°C)  
const float MAX_TEMP = 30.0; // Maximum temperature (°C)  
  
// LDR range for brightness control  
const int MIN_LDR = 100; // Bright condition (lower value = more light)  
const int MAX_LDR = 900; // Dark condition (higher value = less light)  
  
void setup() {  
    Serial.begin(9600);  
    pinMode(FAN_PIN, OUTPUT);  
    pinMode(BULB_PIN, OUTPUT);  
    Serial.println("=== Smart Energy-Saving System Started ===");  
}  
  
void loop() {  
    // Read Temperature from TMP36  
    int tempValue = analogRead(TMP36_PIN);  
    float voltage = tempValue * (5.0 / 1023.0);  
    float temperatureC = (voltage - 0.5) * 100.0;  
  
    // Read Light Level from LDR  
    int ldrValue = analogRead(LDR_PIN);  
  
    // FAN SPEED CONTROL (PWM)
```

```

int fanSpeed = map(temperatureC, MIN_TEMP, MAX_TEMP, 0, 255);
fanSpeed = constrain(fanSpeed, 0, 255);
analogWrite(FAN_PIN, fanSpeed);

// BULB BRIGHTNESS CONTROL (PWM)
int lightBrightness = map(ldrValue, MIN_LDR, MAX_LDR, 0, 255);
lightBrightness = constrain(lightBrightness, 0, 255);
analogWrite(BULB_PIN, lightBrightness);

// Serial Monitor Display
Serial.print("Temperature: ");
Serial.print(temperatureC);
Serial.print(" °C | Fan Speed: ");
Serial.print(fanSpeed);
Serial.print(" | LDR Value: ");
Serial.print(ldrValue);
Serial.print(" | Bulb Brightness: ");
Serial.println(lightBrightness);

delay(500);
}

```

10 Conclusion

The Energy-Saving Fan and Light Automation System successfully demonstrates an intelligent approach to home automation that significantly reduces energy consumption. By integrating simple sensors with an Arduino microcontroller, the system provides automatic control of lighting and cooling appliances based on real-time environmental conditions.

The system effectively addresses the problem of energy wastage by eliminating manual operation and ensuring appliances operate only when needed. Through PWM-based control, it offers smooth transitions in brightness and fan speed, enhancing user comfort while maintaining energy efficiency. The use of low-cost components like TMP36 and LDR makes this solution economically viable for widespread implementation.

Key Achievements

- **Automatic lighting control** based on ambient light conditions using LDR sensor
- **Intelligent temperature regulation** with fan speed control via TMP36 sensor
- **Significant energy savings** through optimized operation and PWM control
- **Cost-effective implementation** using readily available, low-cost components
- **Real-time environmental monitoring** with responsive actuator control
- **Smooth transitions** in brightness and fan speed for enhanced user comfort
- **Practical IoT implementation** creating smart, energy-efficient environments
- **Scalable foundation** for advanced smart home automation systems

Future Enhancements

- Wireless connectivity for remote monitoring and control
- Integration with renewable energy sources like solar power
- Machine learning algorithms for predictive and adaptive control
- Mobile application interface for user customization
- Energy consumption analytics and reporting features

This project demonstrates the potential of embedded systems in creating sustainable and intelligent living spaces while maintaining cost-effectiveness and reliability for real-world deployment. The system serves as a comprehensive solution for modern energy-efficient home automation.

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